

This question paper contains 8 printed pages]

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S. No. of Question Paper : 1135

Unique Paper Code : 235604

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Name of the Paper : Discrete Mathematics

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : VI

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Do any two parts from each question.

Section I

1. (a) Prove that two finite ordered sets P and Q are order isomorphic if and only if they can be drawn with the identical diagrams. 6

- (b) Draw the diagrams of ordered sets

$$2 \times 3, \bar{2} \cup 3, 1 \oplus \bar{3} \oplus 1$$

where n denotes the chain $0 < 1 < \dots < n-1$ and $\bar{n}$ is the anti-chain for the set $\{0, 1, \dots, n-1\}$. 6

P.T.O.

- (c) Find if the mapping $\theta : P \rightarrow Q$ is order preserving for
 $P = \langle \mathcal{P}(S), \subseteq \rangle$ with $|S| > 1$, $Q = 2$ and $\theta(U) = 1$ if
 $U = S$ and $\theta(U) = 0$ if $U \neq S$ where $\mathcal{P}(S)$ is the
 power set of S consisting of all subsets of S .

2. (a) Let $(L, \vee, \wedge)$ be a non-empty set equipped with two
 binary operations which satisfy associative, commutative,
 idempotency and absorption laws. Define a relation ' $\leq$ '
 on L by $a \leq b$ if and only if $a \wedge b = a$. Prove that
 $(L, \leq)$ is a lattice ordered set.

- (b) If ' $\leq$ ' is defined on any subset of $N_0 = N \cup \{0\}$ by
 $m \leq n$ if and only if m divides n then show that this
 ordered subset

$$Q = \{1, 2, 4, 5, 6, 12, 20, 30, 60\}$$

of $(N_0, \leq)$ does not form a lattice. Draw a diagram
 of Q and find elements $a, b, c, d \in Q$ such that $a \vee b$
 and $c \wedge d$ do not exist in Q .

(c) Let L be a lattice. Prove that the following are equivalent :

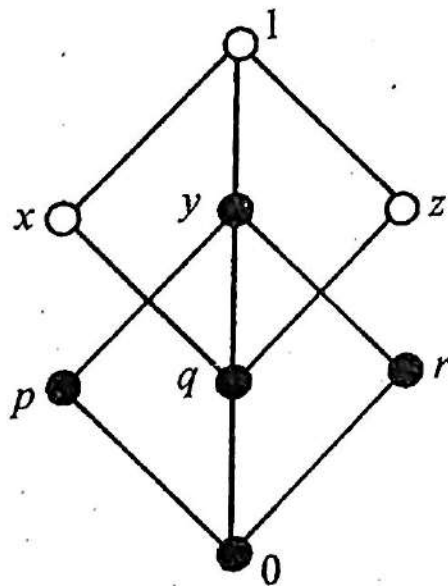
(i) L is a chain.

(ii) Every non-empty subset of L is a sublattice.

(iii) Every two element subset of L is a sublattice. $6\frac{1}{2}$

Section II

3. (a) Define modular and distributive lattices. Use $M_3 - N_5$ theorem to find if the following lattice is modular or distributive. 6



- (b) Find the disjunctive normal form of :

6

$$f(x_1, x_2, x_3) = x_1(x_2 + x_3)' + (x_1x_2 + x_3)x_1.$$

- (c) Find all the prime implicants of :

$$xy'z + x'yz' + xyz' + xyz$$

and form the corresponding prime implicant table. 6

4. (a) (i) In a Boolean algebra B, show that $\forall x, y \in B$

$$x \leq y \Leftrightarrow x \wedge y' = 0 \Leftrightarrow x' \vee y = 1 \Leftrightarrow x \wedge y$$

$$= x \Leftrightarrow x \vee y = y \quad 3\frac{1}{2}$$

- (ii) Show that $(B, \text{gcd}, \text{lcm})$ is a Boolean algebra for the set B of all positive divisors of 30. 3

- (b) Use a Karnaugh map diagram to simplify :

$$(x_1 + x_2)(x_1 + x_3) + x_1x_2x_3. \quad 6\frac{1}{2}$$

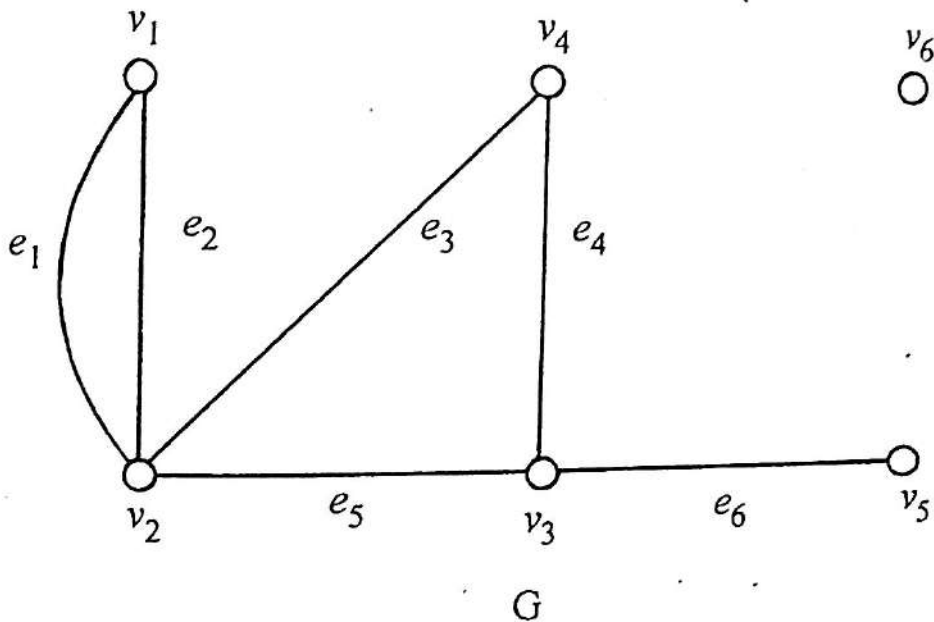
- (c) Draw the symbolic representation of the circuit given by :

$$p = (x_1 + x_3)'(x_1' + (x_2 + x_3)(x_2' + x_3')). \quad 6\frac{1}{2}$$

Section III

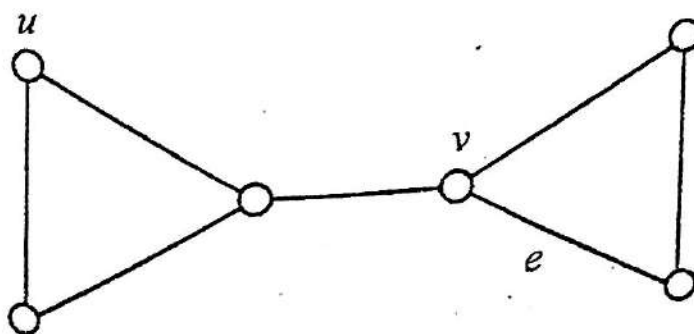
- (a) (i) Find the degree sequence for G . Verify that the sum of the degrees of the vertices is an even number ? Which vertices are even ? Which are odd ?

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- (ii) Define a subgraph of a graph. Draw pictures of the subgraphs $G \setminus \{e\}$, $G \setminus \{v\}$, $G \setminus \{u\}$ where G is :

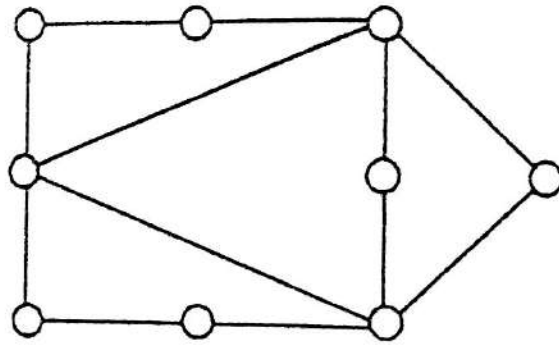
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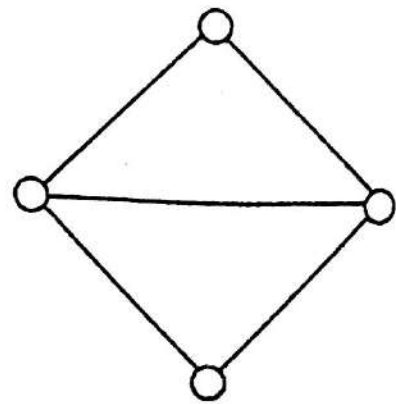
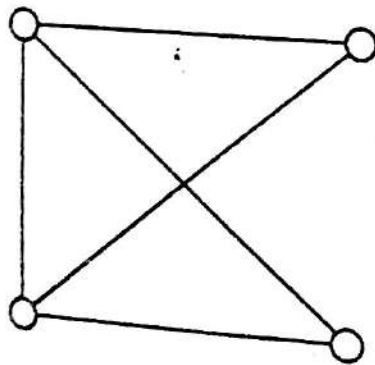
- (b) (i) Determine whether the graph below is bipartite. Give the bipartition sets or explain why the graph is not bipartite.

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- (ii) Define isomorphism of graphs. Also label the graphs so as to show an isomorphism.

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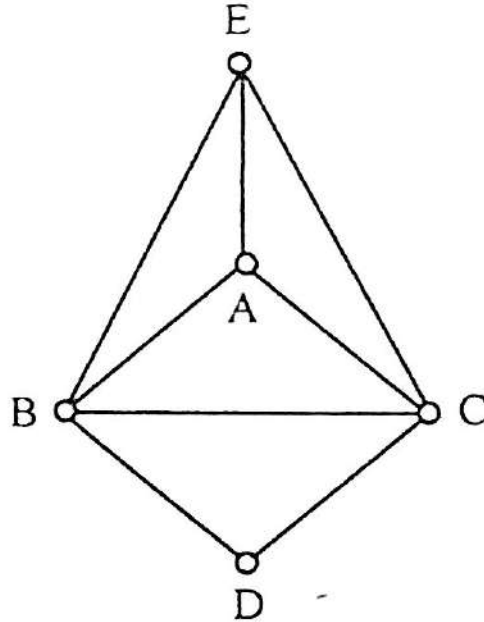


- (c) (i) Define a complete graph. Does there exist a graph G with 28 edges and 12 vertices each of degree 3 or 4. Justify your answer.
- (ii) Why can't there exist a graph whose degree sequence is 5, 4, 4, 3, 2, 1? Draw a graph with degree sequence 4, 3, 2, 2, 1.

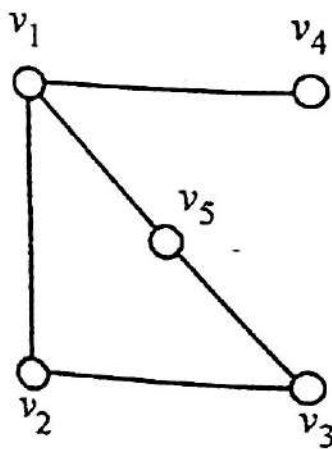
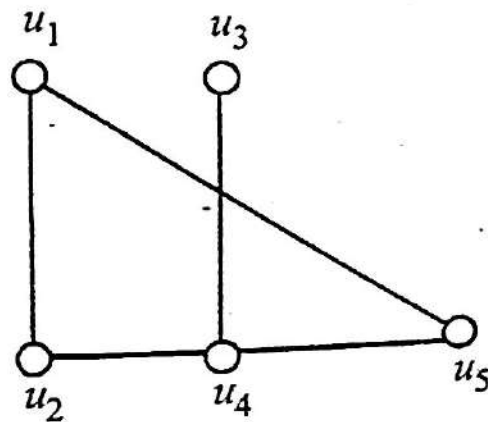
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- (a) Define a Hamiltonian graph. Is the following graph Hamiltonian? Is it Euclidean? Explain your answer. $6\frac{1}{2}$



- (b) Find the adjacency matrices A_1 and A_2 of the graphs G_1 and G_2 shown below. Find a permutation matrix P such that $A_2 = PA_1P^T$. $6\frac{1}{2}$

 G_1  G_2

- (c) Apply the original version of Dijkstra's algorithm to find a shortest path from A to F. Write steps. 6½

